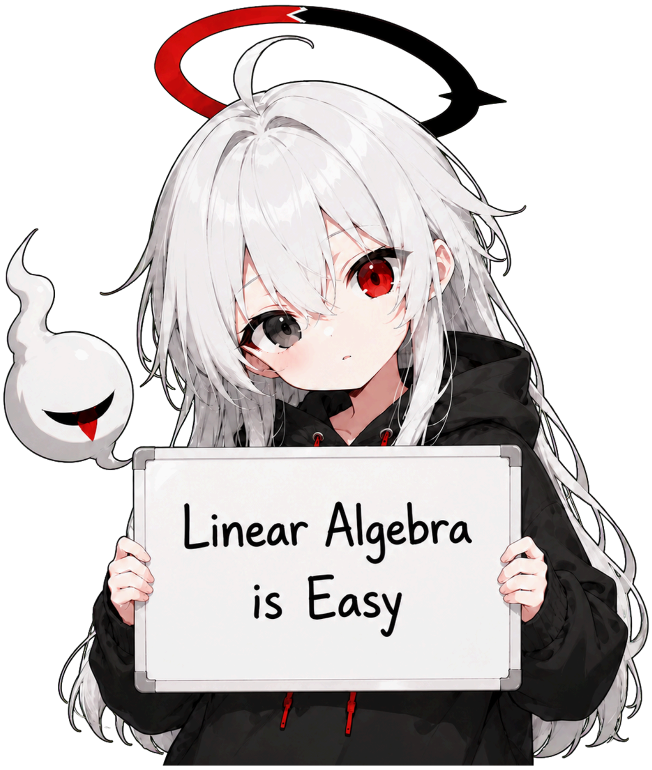
Linear Algebra F*cked Up

Higher Algebra



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Contents

This is my personal notes on Higher Algebra.

0 Preliminaries

0.1 Pointed Simplicial Sets

0.1.1 Definition

Definition 0.1.1 (Pointed Simplicial Sets). The category of *pointed simplicial sets* is defined to be

$$\mathbf{sSet}_* := \mathbf{sSet}_{\Delta^0/} = \left(\Delta^0 \downarrow \mathbf{sSet}\right)$$

where Δ^0 is the obviously the terminal object of **sSet**.

Remark. Since for every n, there is a unique

$$\Delta^n \to \Delta^0$$

the base point of a pointed simplicial set $\Delta^0 \underset{0}{\overset{x}{\to}} x$ induces a constant n-simplex

$$(x_0)_n \in x_n, \quad \Delta^n \to \Delta^0 \to x$$

Geometrically, it is a n-simplex that is "collapsed" to the base point, with all its vertices identified with the base point.

0.1.2 Simplicial n-loops

Just like what we did in topology, we can define some "loops" in simplicial sets.

Definition 0.1.2 (Simplicial n-Loop). A *simplicial n-loop* of a pointed simplicial set (x, x_0) is a simplex $\sigma \in x_n$, where

$$d_i\sigma = (x_0)_{n-1}, \quad i = 0, \ldots, n$$

which means that the border of σ degenerate to the base point.

Remark. Equivalently, the restriction of $\sigma : \Delta^n \to x$ to $\partial\Delta^n$ is the constant map at x_0.

$$\begin{tikzcd}
\partial\Delta^n \arrow[r] \arrow[d] & \Delta^0 \arrow[d, "x_0"] \\
\Delta^n \arrow[r, "\sigma"'] & x
\end{tikzcd}$$

Because the border was collapsed to a point, σ factors uniquely through the quotient

$$\Delta^n/\partial\Delta^n \to x$$

where the notation $\Delta^n/\partial\Delta^n$ is somehow called the *n-sphere* S^n, defined by above. Hence we say

“ simplicial n-loop $\Longleftrightarrow$ pointed simplicial map $S^n \to x$ ”

0.2 Wedge Products and Smash Products

0.2.1 Wedge Products

Definition 0.2.3 (Wedge Product). Suppose $\Delta^0 \to x$ and $\Delta^0 \to y$ are two pointed simplicial sets, the *wedge product* of x and y is defined to be

$$x \vee y := x \sqcup_{\Delta^0} y$$

basically connecting the two base points together.

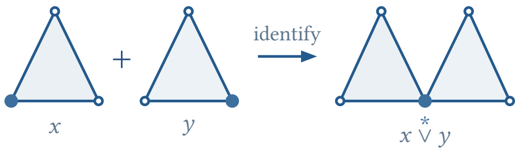

Figure 1: The wedge product identifies the marked base vertices of two pointed simplicial sets.

Remark. The wedge product is functorial in both variables, it is a bifunctor of 1-category

$$(- \vee -) : \mathbf{sSet}_* \times \mathbf{sSet}_* \to \mathbf{sSet}_*$$

also on the category of animae, this is also a ∞-bifunctor

$$(- \vee -) : \mathbf{Ani}_* \times \mathbf{Ani}_* \to \mathbf{Ani}_*$$

0.2.2 Smash Products

Definition 0.2.4 (Smash Product). Suppose $\Delta^0 \to x$ and $\Delta^0 \to y$ are two pointed simplicial sets, the *smash product* of x and y is defined to be

$$x \wedge y := \frac{x \times y}{x \vee y}$$

basically collapsing the wedge product to a point.

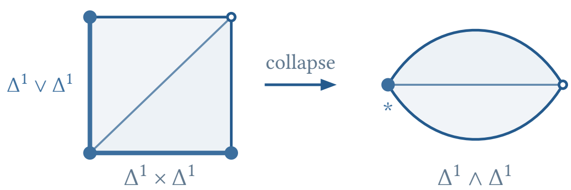

Figure 2: A minimal model of the smash product: the highlighted wedge in the triangulated product is collapsed to the marked vertex.

0.2.3 Some Theories on Wedge and Smash

Before discussing the formal properties, it is useful to keep the following analogy in mind: pointed spaces behave much like modules, with the base point playing the role of zero, the wedge product the role of direct sum, and the smash product the role of tensor product. This is a guide to intuition rather than an identification of the two settings.

R-modules	pointed spaces
0	$*$
$m+n$	$x \vee y$
$m \otimes_R n$	$x \wedge y$
$\mathrm{Hom}_R(m, n)$	$\mathrm{Map}_*(x, y)$

All the equivalences below are natural in the pointed simplicial sets involved. They can equally be read in the ∞-category $\mathbf{Ani}_*$ of pointed animae.

Proposition 0.2.1 (Wedge Product Laws). The wedge product is symmetric and associative, and the one-point object $*$ is its unit:

$$x \vee y \simeq y \vee x$$
$$(x \vee y) \vee z \simeq x \vee (y \vee z)$$
$$x \vee * \simeq x$$

These equivalences express the fact that $x \vee y$ is the coproduct of x and y in the pointed category.

Proposition 0.2.2 (Smash Product Laws). Let $S^0 = \Delta^0 \sqcup \Delta^0$, with one summand chosen as the base point. The smash product is symmetric and associative, its unit is S^0, and the one-point object is absorbing:

$$x \wedge y \simeq y \wedge x$$
$$(x \wedge y) \wedge z \simeq x \wedge (y \wedge z)$$
$$x \wedge S^0 \simeq x$$
$$x \wedge * \simeq *$$

Proposition 0.2.3 (Distributivity and Tensor-Hom Adjunction). Smash product distributes over wedge product:

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z).$$

Moreover, pointed mapping spaces satisfy the natural adjunction

$$\mathrm{Map}_*(x \wedge y, z) \simeq \mathrm{Map}_*(x, \mathrm{Map}_*(y, z)).$$

Thus $x \wedge (-)$ preserves colimits, which explains the distributive law.

Proof. Write x_0, y_0, and z_0 for the chosen base points. By definition, the quotient map

$$q : x \times y \to x \wedge y$$

collapses $x \vee y = (x \times \{y_0\}) \cup (\{x_0\} \times y)$ to the base point. Hence a pointed map $g : x \wedge y \to z$ is the same thing as a map $f : x \times y \to z$ satisfying

$$f(x, y_0) = z_0 = f(x_0, y)$$

for every x and y.

Currying f gives a map

$$x \to \mathrm{Map}(y, z), \quad x \mapsto (y \mapsto f(x, y))$$

The condition $f(x, y_0) = z_0$ says that every map $y \to z$ in its image is pointed, so this map factors through $\mathrm{Map}_*(y, z)$. The condition $f(x_0, y) = z_0$ says that x_0 is sent to the constant base-point map, so the curried map $x \to \mathrm{Map}_*(y, z)$ is itself pointed. This construction is reversible and natural. Applied degreewise, it gives the equivalence of pointed mapping spaces

$$\mathrm{Map}_*(x \wedge y, z) \simeq \mathrm{Map}_*(x, \mathrm{Map}_*(y, z))$$

By symmetry, $x \wedge (-)$ is therefore left adjoint to $\mathrm{Map}_*(x, -)$ and preserves all colimits. Since $y \vee z$ is the coproduct of y and z in the pointed category, preservation of this coproduct gives

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z)$$

as claimed. □

In other words, $(\mathbf{sSet}_*, \wedge, S^0)$ is a closed symmetric monoidal category, with internal Hom object $\mathrm{Map}_*(-, -)$. Likewise, $(\mathbf{Ani}_*, \wedge, S^0)$ is a closed symmetric monoidal ∞-category.

> **Remark (Spheres and Suspension).** For pointed spheres one obtains
>
> $$S^m \wedge S^n \simeq S^{m+n}$$
>
> the proof is obvious and direct.

0.3 Suspension and Looping

0.3.1 Suspension

Intuitively, the suspension of a pointed simplicial set is obtained by taking the product with a simplicial interval and collapsing the two ends to points. Formally, we have

> **Definition 0.3.5 (Suspension).** The *suspension* of a pointed simplicial set x is defined to be
>
> $$\Sigma x := S^1 \wedge x$$

0.3.2 Looping

According to the "n-loop = pointed map from n-sphere" principle, we have

> **Definition 0.3.6 (Loop).** The *loop* of a pointed simplicial set x is defined to be
>
> $$\Omega x := \mathrm{Map}_*(S^1, x)$$

0.3.3 Some theories on Suspension and Looping

Theorem 0.3.1 (Suspension-Loop Adjunction). There is a natural equivalence of pointed mapping spaces

$$\mathrm{Map}_*(\Sigma x, y) \simeq \mathrm{Map}_*(x, \Omega y)$$

for every pointed simplicial sets x and y. In other words, there is an adjunction

$$\Sigma : \mathbf{sSet}_* \rightleftarrows \mathbf{sSet}_* : \Omega$$

Proof. Immediately by definition and by Proposition 0.2.3. □

0.3.4 Module Analogy

Let R be a commutative ring. For a fixed R-module ℓ, the ordinary tensor–Hom adjunction is

$$\mathrm{Hom}_R(\ell \otimes_R m, n) \simeq \mathrm{Hom}_R(m, \mathrm{Hom}_R(\ell, n)).$$

Thus the pointed-space dictionary extends from individual operations to adjoint functors:

$$\begin{array}{ccc} k \wedge (-) & \leftrightarrow & \ell \otimes_R (-) \\ \mathrm{Map}_*(k, -) & \leftrightarrow & \mathrm{Hom}_R(\ell, -). \end{array}$$

Taking $k = S^1$ on the space side gives $\Sigma \dashv \Omega$. The closer algebraic analogue appears in the derived category $D(R)$, where the shifted unit $R[1]$ satisfies

$$\begin{array}{c} R[1] \otimes^{\mathrm{L}}_R m \simeq m[1] \\ \mathrm{RHom}_R(R[1], n) \simeq n[-1]. \end{array}$$

Hence the shift $[1]$ is left adjoint to $[-1]$, and the two are inverse equivalences. This last feature belongs to the stable setting: after pointed spaces are stabilized to spectra, Σ and Ω likewise become inverse equivalences; in $\mathbf{Ani}_*$ they are adjoint but not inverse in general. We can even claim ambitiously

> “Stable homotopy theory = Linear algebra over the sphere spectrum $\mathbb{S}$.”

We will explore this analogy in further chapters.

0.4 Homotopy Groups

The simplicial n-loops introduced earlier are individual representatives. To obtain an invariant of the homotopy type of x, we identify representatives that are homotopic through pointed maps.

Definition 0.4.7 (Homotopy Groups). Let (x, x_0) be a pointed anima, represented for example by a pointed Kan complex. For every $n \geq 0$, define

$$\pi_n(x, x_0) := \pi_0 \mathrm{Map}_*(S^n, x) = [S^n, x]_*,$$

where $[S^n, x]_*$ denotes the set of pointed homotopy classes of maps. In terms of iterated loop spaces, the same definition is

$$\pi_n(x, x_0) = \pi_0(\Omega^n x),$$

where $\Omega^0 x = x$ and the base point of $\Omega^n x$ is the constant loop at x_0.

Remark (Why a Kan Replacement Is Needed). If x is an arbitrary pointed simplicial set, the expression above means the *derived* pointed mapping space. Concretely, choose a pointed Kan fibrant replacement $x \to x^{\mathrm{fib}}$ and set

$$\pi_n(x, x_0) := \pi_0 \mathrm{Map}_*\left(S^n, x^{\mathrm{fib}}\right).$$

This makes π_n invariant under weak equivalence. When x is already Kan, no replacement is necessary.

Remark (Simplicial Description). For a pointed Kan complex (x, x_0), a map $S^n = \Delta^n/\partial\Delta^n \to x$ is exactly an n-simplex $\sigma \in x_n$ satisfying

$$d_i\sigma = (x_0)_{n-1}, \quad 0 \le i \le n.$$

Therefore $\pi_n(x, x_0)$ is the set of such simplicial n-loops modulo pointed simplicial homotopy relative to $\partial\Delta^n$. The quotient by homotopy is essential: the raw set of n-loops is not itself a homotopy invariant.

Proposition 0.4.4 (Group Structure). For $n \ge 1$, $\pi_n(x, x_0)$ is a group; for $n \ge 2$, it is abelian. The product of two classes $[f]$ and $[g]$ is represented by the composite

$$S^n \xrightarrow{\text{pinch}} S^n \vee S^n \xrightarrow{f \vee g} x.$$

Equivalently, this is concatenation of components in the n-fold loop space $\Omega^n x$. The identity is the constant map at x_0.

Proof sketch. The loop space $\Omega^n x$ has concatenation, so its connected components form a group when $n \ge 1$. For $n \ge 2$, the two loop directions give two compatible multiplications; the Eckmann–Hilton argument makes the product commutative. □

0.4.1 The Whitehead Theorem

Theorem 0.4.2 (Whitehead Theorem). Suppose $f : x \to y$ is a map of animae. Suppose

$$\pi_0(x) \xrightarrow{\sim} \pi_0(y), \quad \pi_n(x, x_0) \xrightarrow{\sim} \pi_n(y, f(x_0))$$

holds for every base point x_0. Then f is a categorical equivalence.

1 Stable ∞-Categories

1.1 The Motivation of the Subject

Ordinary category theory remembers a *set* of maps between two objects. This is often enough in classical algebra, but it is too small for homotopy theory and derived mathematics. There, two maps may be connected by a homotopy, two homotopies may themselves be homotopic, and these higher relations affect later constructions. Higher Algebra keeps this information instead of collapsing it.

Localization. When quasi-isomorphisms or weak equivalences are inverted, an ordinary localization remembers only homotopy classes of maps. An ∞-categorical localization retains the whole mapping space. Its truncation is

$$\mathrm{Hom}_{hC}(x, y) = \pi_0 \mathrm{Map}_C(x, y)$$

Thus the ordinary derived category is useful, but it is only a shadow of the enhanced object.

Coherence. "Associative up to homotopy" is not enough by itself. The chosen homotopies must agree with one another, their agreements must also agree, and so on. $\mathbb{A}_\infty$- and $\mathbb{E}_n$-algebras, encoded by ∞-operads, package this infinite system of compatibility conditions.

Stability. Stable ∞-categories provide functorial fibers, cofibers, shifts, and mapping spectra. They unify derived categories and spectra while retaining the information that a triangulated homotopy category forgets.

> Remark (Worldview). Higher Algebra replaces "unique by equality" with "a contractible space of coherent choices." Its guiding principle is simple: keep enough structure so that the next construction is invariant, functorial, and composable.

In short, ∞-categories retain higher morphisms, stable ∞-categories handle derived and exact phenomena, and ∞-operads describe algebraic operations up to every required coherence. These tools are used because the mathematics is already higher, not because abstraction is an end in itself.

1.2 First Definitions: Stability

1.2.1 Pointed ∞-Categories

> **Definition 1.2.1 (Zero Object, Pointed ∞-Category).** Let C be an ∞-category, a *zero object* is an object which is both initial and terminal. We will say that C is *pointed* if it admits a zero object. In this case, we will denote the zero object by $0 \in C$.

Obviously, if C is pointed, then the zero object is unique up to a contractible space of choices. The condition is equals to:

Proposition 1.2.1. Let C be an ∞-category, C is pointed iff:

1. C admits initial and terminal objects $0, 1$.
2. Exists $0 \to 1$.

1.2.2 Triangle, Fiber and Cofiber

Definition 1.2.2 (Triangle, (co)Fiber Sequence). Let C be a pointed ∞-category, a *triangle* in C is a diagram $\Delta^1 \times \Delta^1 \to C$ depicted as

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d] & y \arrow[d, "g"] \\ 0 \arrow[r] & z \end{tikzcd}$$

where 0 is the zero object. We say that a triangle in C is a *fiber sequence* if it is a pullback square, and a *cofiber sequence* if it is a pushout square.

A triangle is basically a composite of two maps

$$x \xrightarrow{f} y \xrightarrow{g} z, \quad gf \simeq 0$$

if this triangle is a pullback, then $x \simeq y \times_z 0$, where $y \times_z 0$ is exactly the fiber of g, hence $x \simeq \mathrm{fib}(g)$. Similarly, if this triangle is a pushout, then $z \simeq y \sqcup_x 0$, where $y \sqcup_x 0$ is exactly the cofiber of f, hence $z \simeq \mathrm{cofib}(f)$.

Remark. Let C be a pointed ∞-category, a triangle in C consists of the following data:

1. A pair of morphisms $f : x \to y$ and $g : y \to z$ in C.
2. A 2-simplex in C corresponding to a diagram

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & {} \arrow[u, phantom, "///"] & z \end{tikzcd}$$

in C, which witnesses h as a composite of f and g.

3. A 2-simplex in C corresponding to a diagram

$$\begin{tikzcd} & 0 \arrow[dr, "0"] & \\ x \arrow[ur, "0"] \arrow[rr, "h"'] & {} \arrow[u, phantom, "///"] & z \end{tikzcd}$$

in C, which may view as a nullhomotopy of h.

By the notion of triangle, we can define the notion of fiber and cofiber in a pointed ∞-category.

Definition 1.2.3 ((co)Fiber). Let C be a pointed ∞-category containing a morphism $g : x \to y$, a *fiber* of g is a fiber sequence

$$\begin{tikzcd} w \arrow[r] \arrow[d] & x \arrow[d, "g"] \\ 0 \arrow[r] & y \end{tikzcd}$$

Dually, a *cofiber* of g is a cofiber sequence

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[d] & y \arrow[d] \\ 0 \arrow[r] & z \end{tikzcd}$$

We write $w = \mathrm{fib}(g)$ and $z = \mathrm{cofib}(g)$.

Proposition 1.2.2. Fix a pointed ∞-category C and a morphism $f : x \to y$, if a cofiber of f exists, then it is unique up to an equivalence. Precisely, consider the full subcategory

$$E \subset \mathrm{Fun}(\Delta^1 \times \Delta^1, C)$$

spanned by the cofiber sequences, let $\theta : E \to \mathrm{Fun}(\Delta^1, C)$ be the forgetful functor, which send a diagram

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[d] & y \arrow[d] \\ 0 \arrow[r] & z \end{tikzcd}$$

to the morphism $g : x \to y$. Then θ is a Kan fibration. If every morphism in C admits a cofiber, then θ is a trivial Kan fibration.

Proposition 1.2.3. The functor

$$\mathrm{cofib} : \mathrm{Fun}(\Delta^1, C) \to C$$

is the left adjoint of the follwing functor:

$$C \simeq \mathrm{Fun}(\Delta^0, C) \to \mathrm{Fun}(\Delta^1, C)$$

by sending every object x to the morphism $0 \to x$. Hence cofib preserves colimits.

1.2.3 Stability

Definition 1.2.4 (Stable ∞-Category). An ∞-category C is *stable* if it satisfies the following conditions:

1. C is pointed.
2. Every morphism in C admits a fiber and a cofiber.
3. A triangle in C is a fiber sequence if and only if it is a cofiber sequence.

Proposition 1.2.4. If C is a stable ∞-category, then C^{op} is also stable.

1.2.4 Closure Properties

Proposition 1.2.5. Let C be a stable ∞-category and k a simplicial set. Then the ∞-category $\mathrm{Fun}(k, C)$ is stable.

Proposition 1.2.6. Let C be a small stable ∞-category, and let κ be a regular cardinal, then the ∞-category $\mathrm{Ind}_\kappa(C)$ is stable.

1.3 Homotopy Category of a Stable ∞-Category

1.3.1 Intro

Let m be a module over a commutative ring R. It admits a resolution

$$\ldots \to p_2 \to p_1 \to p_0 \to m \to 0$$

by projective R-modules p_i. There are generally many different choices, but they are always *quasi-isomorphic* to each other: that is, a chain map between two resolutions that induces a isomorphism on homology. It's often convenient to work in the *derived category* $\mathrm{D}(R)$ of the ring R, the category obtained from the category of chain complexes of R-modules by formally inverting quasi-isomorphisms.

The derived category $\mathrm{D}(R)$ of a commutative ring R is usually not an abelian category. A morphism $f : x' \to x$ in $\mathrm{D}(R)$ usually does not have a kernel in $\mathrm{D}(R)$. Instead, one can associate to f its cofiber (*mapping cone*) x'', which is well-defined up to noncanonical isomorphism.

Verdier introduced the notion of a *triangulated category* to axiomatize the properties of the derived category by formating the mapping cones, we will soon review the theory of triangulated categories and then show that the homotopy category of a stable ∞-category is triangulated.

1.3.2 Basic Homological Algebra

Definition 1.3.5 (Additive Category, Abelian Category). Let A be a category, we say that A is *additive* if it satisfies the following conditions:

1. A admits a zero object 0.
2. A admits finite products and coproducts.

For every pair of objects $x, y \in A$, a *zero morphism* is a map factors through $x \to 0 \to y$. It is unique, and denoted by 0.

3. For every pair x, y the map $x \sqcup y \to x \times y$ described by

$$\begin{pmatrix} \mathrm{id}_x & 0 \\ 0 & \mathrm{id}_y \end{pmatrix}$$

   is an isomorphism, and denote the inverse by $\varphi_{x,y}$.

We can now define the *sum* of two morphisms $f, g : x \to y$ to be the morphism $f + g : x \to y$ by the composition

$$x \xrightarrow{\Delta_x} x \times x \xrightarrow{f \times g} y \times y \xrightarrow{\varphi_{y,y}} y \sqcup y \xrightarrow{\nabla_y} y$$

where Δ_x is the diagonal map and ∇_y is the codiagonal map. This construction makes $\mathrm{Hom}_A(x, y)$ into a commutative monoid with identity 0.

4. For every x, y the commutative monoid $\mathrm{Hom}_A(x, y)$ is an abelian group. i.e. every morphism $f : x \to y$ admits an additive inverse $-f : x \to y$ such that $f + (-f) = 0$.

An additive category A is *abelian*, if every morphism admits a kernel and a cokernel, and every monomorphism is a kernel of its cokernel, and the canonical map

$$\mathrm{coker}(\ker(f) \to x) \to \ker(y \to \mathrm{coker}(f))$$

is an isomorphism for every morphism $f : x \to y$ in A.

Definition 1.3.6 (Triangulated Category). A *triangulated category* consists of the following data:

1. An additive category A.
2. A translation functor $A \to A$ which is an equivalence of categories, by $x \mapsto x[1]$.
3. A collection of *distinguished triangles*

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

These data are required to satisfy the following axioms:

1. Every morphism $f : x \to y$ in A can be extended to distinguished triangle in A.
2. The collection of distinguished triangles is stable under isomorphisms.
3. Given an object $x \in A$, the diagram

$$x \xrightarrow{\mathrm{id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

   is a distinguished triangle.
4. A diagram

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

   is a distinguished triangle if and only if the diagram

$$y \xrightarrow{g} z \xrightarrow{h} x[1] \xrightarrow{-f[1]} y[1]$$

   is a distinguished triangle.
5. Given a commutative diagram

$$
\begin{tikzcd}
x \arrow[r] \arrow[d, "f"'] & y \arrow[r] \arrow[d] & z \arrow[r] \arrow[d, dashed] & x[1] \arrow[d, "f[1]"] \\
x' \arrow[r] & y' \arrow[r] & z' \arrow[r] & x'[1]
\end{tikzcd}
$$

where the rows are distinguished triangles, then there exists a morphism $z \to z'$ making the whole diagram commutative.

6. (*Octahedral Axiom*) uppose given three distinguished triangles

$$x \xrightarrow{f} y \xrightarrow{u} y/x \xrightarrow{d} x[1], \quad y \xrightarrow{g} z \xrightarrow{v} z/y \xrightarrow{d'} y[1], \quad x \xrightarrow{g\circ f} z \xrightarrow{w} z/x \xrightarrow{d'''} x[1]$$

in A, there exists a fourth distinguished triangle

$$y/x \xrightarrow{\varphi} z/x \xrightarrow{\psi} z/y \xrightarrow{\theta} y/x[1]$$

such that the diagram

$$
\begin{tikzcd}
 & z \arrow[dr, "w"] \arrow[ddr, "v"] & \\
x \arrow[ur, "g \circ f"] \arrow[d, "f"'] & & z/x \arrow[ll, "{d'', +1}"] \arrow[d, "\psi"] \\
y \arrow[uur, "g"] \arrow[dr, "u"'] & & z/y \arrow[ll, "{d', +1}"] \arrow[dl, "{\theta, +1}"] \\
 & y/x \arrow[uul, "{d, +1}"] \arrow[uur, "\varphi"] &
\end{tikzcd}
$$

commutes.

1.3.3 Homotopy Category of a Stable ∞-Category

Let C be a stable ∞-category. The entire construction rests on two facts: pushout squares are pullback squares, and the space of choices of a limit or colimit is contractible. The first produces the usual homological operations; the second makes them coherent.

1.3.3.1 Suspension and Additivity

Definition 1.3.7 (Suspension and Loop Objects). For $x \in C$, define Σx and Ωx by a pushout square and a pullback square, respectively:

$$
\begin{tikzcd}
x \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \Sigma x
\end{tikzcd}
\qquad
\begin{tikzcd}
\Omega x \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & x
\end{tikzcd}
$$

> The left square is a pushout and the right square is a pullback. We write $x[1] = \Sigma x$ and $x[-1] = \Omega x$

> **Proposition 1.3.7 (Suspension Is an Equivalence).** The functors $\Sigma, \Omega : C \to C$ are inverse equivalences. Hence $[1] : hC \to hC$ is an equivalence.

Proof. The square defining Σx is also a pullback, so $\Omega\Sigma x \simeq x$. Dually, $\Sigma\Omega x \simeq x$. These equivalences are natural because the relevant spaces of choices are contractible. □

> **Proposition 1.3.8 (Additivity of the Homotopy Category).** The homotopy category hC is additive.

Proof. The zero map points every $\mathrm{Map}_C(x, y)$, and suspension gives

$$\mathrm{Map}_C(\Sigma x, y) \simeq \Omega \mathrm{Map}_C(x, y)$$

Therefore

$$\mathrm{Hom}_{hC}(x, y) \simeq \pi_2 \mathrm{Map}_C(x[-2], y)$$

The right side is an abelian group, and naturality makes composition bilinear. Thus hC is preadditive.

The zero object remains a zero object in hC. For $x, y \in C$, put

$$p := \mathrm{cofib}\left(x[-1] \xrightarrow{0} y\right)$$

The pushout universal property and the suspension equivalence give

$$\mathrm{Map}_C(p, t) \simeq \mathrm{Map}_C(x, t) \times \mathrm{Map}_C(y, t)$$

Hence p is a coproduct. A finite coproduct in a preadditive category is a biproduct: its projections are determined by the matrices $(\mathrm{id}, 0)$ and $(0, \mathrm{id})$. Thus hC is additive. □

1.3.3.2 Distinguished Triangles

For a morphism $f : x \to y$, choose a coherent diagram in C

$$\begin{tikzcd}
x \arrow[r, "f"] \arrow[d] & y \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & C_f \arrow[r, "\delta_f"] & \Sigma x
\end{tikzcd}$$

in which both small squares are pushouts. Then $C_f \simeq \mathrm{cofib}\,(f)$, the outer rectangle gives Σx, and δ_f is the connecting morphism.

> **Definition 1.3.8 (Distinguished Triangle in hC).** A triangle in hC
>
> $$x \xrightarrow{f} y \xrightarrow{u} z \xrightarrow{\delta} x[1]$$

is *distinguished* if it is isomorphic to a cofiber triangle

$$x \xrightarrow{f} y \to C_f \xrightarrow[f]{\delta} \Sigma x$$

arising from a coherent two-pushout diagram as above.

Restriction from coherent two-pushout diagrams to the arrow f is a trivial Kan fibration. Consequently its fibers are contractible: the triangle is independent of all choices up to isomorphism in hC, and coherent squares of arrows lift to morphisms of cofiber triangles.

Lemma 1.3.1 (The Sign from Transposing a Suspension Square). Interchanging the two zero directions in a pushout square defining Σx induces $-\mathrm{id}_{\Sigma x}$ in hC.

Proof. Under the natural identification

$$\mathrm{Hom}_{hC}(\Sigma x, y) \simeq \pi_1 \mathrm{Map}_C(x, y)$$

transposition is loop reversal, hence multiplication by -1. Yoneda gives the result. □

Theorem 1.3.1 (Triangulated Homotopy Category). Let C be a stable ∞-category. With translation $x[1] = \Sigma x$ and the distinguished triangles defined above, hC is a triangulated category.

Proof. We verify Verdier's axioms.

TR1: existence, invariance, and the identity triangle. Every arrow $f : x \to y$ extends to

$$x \xrightarrow{f} y \longrightarrow C_f \xrightarrow[f]{\delta} x[1]$$

by two pushouts. Isomorphism invariance is part of the definition, and $C_{\mathrm{id}_x} \simeq 0$ gives

$$x \xrightarrow{\mathrm{id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

TR2: rotation. The right pushout square says that $x[1]$ is the cofiber of $y \to C_f$. One further pushout gives

$$y \xrightarrow{u} C_f \xrightarrow[f]{\delta} x[1] \xrightarrow{-f[1]} y[1]$$

The minus sign is precisely the sign of the preceding lemma. Since shifts preserve pushouts, desuspension and repeated rotation prove the converse.

TR3: morphisms of triangles. A commutative square in hC is a coherent square in C after choosing representatives and a homotopy. The trivial Kan fibration of cofiber extensions lifts it to a morphism of the two cofiber triangles. Its third component $C_f \to C_{f'}$ is the required map; uniqueness is neither asserted nor needed.

TR4: the octahedral axiom. For $x \xrightarrow{f} y \xrightarrow{g} z$, choose coherent cofiber models. Pushout pasting gives

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[d] & y \arrow[r, "g"] \arrow[d] & z \arrow[d] \\ 0 \arrow[r] & C_f \arrow[r, "\varphi"] & C_{gf} \end{tikzcd}$$

The left square and outer rectangle are pushouts, hence so is the right square. Pushing it out along $C_f \to 0$ identifies its cofiber with $C_g = z \sqcup_y 0$ and produces the fourth distinguished triangle

$$C_f \xrightarrow{\varphi} C_{gf} \xrightarrow{\psi} C_g \xrightarrow{\theta} C_{f[1]}$$

All four triangles are faces of one coherent pushout diagram. Its remaining faces give exactly the commutative octahedron, including the shifted signs fixed by TR2. Thus TR4 holds and hC is triangulated. □

Corollary 1.3.1 (The Stable–Triangulated Dictionary). Passing from C to hC carries the basic stable constructions to the usual triangulated ones:

stable ∞-category	triangulated category
zero object	zero object
Σ, Ω	shifts $[1], [-1]$
finite product = finite coproduct	biproduct
cofiber/fiber sequence	distinguished triangle
exact functor	triangulated functor

Thus every formal theorem about triangulated categories applies to hC. Exact functors descend to triangulated functors, and mapping spaces give the usual long exact sequences. The enhancement C retains more: its cofibers and comparison maps are coherently functorial, unlike cones in hC.

1.4 Exact Functors

1.4.1 The Definition

Definition 1.4.9 (Exact Functor). A functor $F : C \to D$ between stable ∞-categories is *exact* if it preserves zero objects and sends cofiber sequences in C to cofiber sequences in D. Equivalently, F preserves fiber sequences.

We have the following equivalent characterizations of exact functors.

Proposition 1.4.9. Let $F : C \to D$ be a functor between stable ∞-categories, then the following are equivalent:

1. F is exact.

2. F preserves finite limits.
3. F preserves finite colimits.

Proof sketch. Suppose first that F is exact. It preserves suspensions, desuspensions, and finite biproducts. Moreover, the pushout of a span $y \xleftarrow{f} x \xrightarrow{g} z$ is the cofiber of

$$x \xrightarrow{(f,-g)} y + z$$

Hence F preserves pushouts. Since it also preserves the zero object, it preserves all finite colimits. Dually, or by expressing a pullback as the fiber of $y + z \to w$, it preserves all finite limits.

Conversely, a functor preserving finite colimits preserves the initial object and pushout squares, hence the zero object and cofiber sequences. A functor preserving finite limits preserves the terminal object and pullback squares; in a stable ∞-category these are again the zero object and cofiber sequences. Thus either condition implies exactness. □

1.4.2 Ext Functor

Definition 1.4.10 (Ext Functor). We write

$$\mathrm{Ext}^n_C(x, y) := \mathrm{Hom}_{hC}(x, y[n])$$

2 Homological Algebra

2.1 t-Structures

2.1.1 Definition of t-Structures on Triangulated Categories

Recall that we say a full sub ∞-category $C' \subset C$ is a *localization* if the inclusion functor has a left adjoint. We will introduce a class of localizations called t-*structures* which are particularly important in homological algebra.

Definition 2.1.1 (t-Structure on Triangulated Category). Let D be a triangulated category. A t-*structure* on D is a pair of full subcategories $(D_{\geq 0}, D_{\leq 0})$ stable under isomorphisms, such that the following conditions hold:

1. For $x \in D_{\geq 0}$ and $y \in D_{\leq 0}$, we have $\mathrm{Hom}_D(x, y[-1]) = 0$.
2. We have inclusions $D_{\geq 0}[1] \subset D_{\geq 0}$ and $D_{\leq 0}[-1] \subset D_{\leq 0}$.
3. For any $x \in D$, there exists a fiber sequence $x' \to x \to x''$ (i.e. $x' = \mathrm{fib}\,(x \to x'')$) where $x' \in D_{\geq 0}$ and $x'' \in D_{\leq -1}$.

Remark. We denote $D_{\geq n}$ for $D_{\geq 0}[n]$ and $D_{\leq n}$ for $D_{\leq 0}[n]$ for any integer n.

Remark. In this definition, either of the full subcategories $D_{\geq 0}$ and $D_{\leq 0}$ determines the other. For example, an object $y \in D$ belongs to $D_{\leq -1}$ if and only if $\mathrm{Hom}_D(x, y)$ vanishes for all $x \in D_{\geq 0}$.

2.1.2 Definition of t-Structures on Stable ∞-Categories

Definition 2.1.2 (t-Structure on Stable ∞-Category). Let C be a stable ∞-category, a t-*structure* on C is a t-structure hC, if C is equipped with a t-structure, we let $C_{\geq n}$ and $C_{\leq n}$ denote the full subcategories of C spanned by those objects which belong to $(hC)_{\geq n}$ and $(hC)_{\leq n}$, respectively.

Proposition 2.1.1. Let C be a stable ∞-category with a t-structure, for each $n \in \mathbb{Z}$, the full subcategory $C_{\leq n}$ is a localization of C.

Proof. Since shifts are equivalences and carry $C_{\leq n}$ to translates of $C_{\leq -1}$, it is enough to treat $n = -1$.

Recall the criterion for a full subcategory to be a localization. The inclusion $i : D \to C$ has a left adjoint if, for every $x \in C$, there is a map $\eta_x : x \to Lx$ with $Lx \in D$ such that composition with η_x induces an equivalence

$$\mathrm{Map}_C(Lx, y) \to \mathrm{Map}_C(x, y)$$

for every $y \in D$. Indeed, this is exactly the universal property characterizing Lx as the reflection of x into D; these universal objects assemble into a functor $L : C \to D$ left adjoint to i.

Apply the third axiom of the t-structure to x. It gives a fiber, hence also cofiber, sequence

$$x' \to x \underset{x}{\overset{\eta}{\to}} x''$$

with $x' \in C_{\geq 0}$ and $x'' \in C_{\leq -1}$. We claim that x'' has the required universal property. Fix $y \in C_{\leq -1}$. Applying $\mathrm{Map}_C(-, y)$ gives a fiber sequence

$$\mathrm{Map}_C(x'', y) \to \mathrm{Map}_C(x, y) \to \mathrm{Map}_C(x', y)$$

It therefore remains to show that $\mathrm{Map}_C(x', y)$ is contractible.

For $k \in \mathbb{Z}$, write

$$\mathrm{Ext}^k_C(a, b) := \mathrm{Hom}_{hC}(a, b[k]) \simeq \mathrm{Hom}_{hC}(a[-k], b)$$

If $k \leq 0$, then $-k \geq 0$, so the shift axiom gives $x'[-k] \in C_{\geq 0}$. The orthogonality axiom and $y \in C_{\leq -1}$ now imply

$$\mathrm{Ext}^k_C(x', y) \simeq \mathrm{Hom}_{hC}(x'[-k], y) \simeq 0$$

For every $m \geq 0$, stability identifies the homotopy groups of the mapping space with these groups:

$$\pi_m \mathrm{Map}_C(x', y) \simeq \mathrm{Ext}^{-m}_C(x', y) \simeq 0$$

Mapping spaces are Kan complexes. The case $m = 0$ says that this mapping space is connected, and the higher cases say that all its homotopy groups vanish; the Whitehead criterion therefore makes it contractible. Hence $\mathrm{Map}_C(x'', y) \to \mathrm{Map}_C(x, y)$ is an equivalence for every $y \in C_{\leq -1}$, so $x \mapsto x''$ defines the left adjoint $\tau_{\leq -1} : C \to C_{\leq -1}$. Shifting gives the localization $\tau_{\leq n} : C \to C_{\leq n}$ for every $n \in \mathbb{Z}$. □

Corollary 2.1.1. Each $C_{\leq n}$ is closed under all limits which exist in C. Dually, each $C_{\geq n}$ is closed under all colimits which exist in C.

2.2 Truncation Functors

2.2.1 Definition of Truncation Functors

Indeed, local objects are characterized by equivalences of mapping spaces, and such equivalences are preserved under limits. The second statement is dual.

Remark (Notation). Write

$$\tau_{\leq n} : C \to C_{\leq n}, \quad \tau_{\geq n} : C \to C_{\geq n}$$

for the left and right adjoints to the respective inclusions. Every $x \in C$ then fits naturally into a fiber–cofiber sequence

$$\tau_{\geq n}x \to x \to \tau_{\leq n-1}x$$

Both truncation functors preserve every $C_{\leq m}$ and every $C_{\geq m}$. For example, $\tau_{\leq n}$ either acts as the identity on $C_{\leq m}$ or lands in the smaller subcategory $C_{\leq n}$. The assertion for $\tau_{\geq n}$ follows from the displayed fiber sequence and closure under limits. The remaining statements are dual.

Remark (Warning). The symbol $\tau_{\leq n}$ here denotes truncation for a t-structure, not the truncation of an arbitrary ∞-category by homotopy dimension. A nonzero object of a stable ∞-category is never truncated in the latter sense: for arbitrarily large r, its identity gives a nonzero class in $\pi_r \mathrm{Map}_C(x[-r], x)$.

The notions agree after restricting the source to $C_{\geq 0}$. For $k \geq -1$, an object x belongs to $C_{\leq k}$ exactly when $\mathrm{Map}_C(y, x)$ is k-truncated for every $y \in C_{\geq 0}$.

2.2.2 Commuting Truncations and the Heart

For $m, n \in \mathbb{Z}$, the universal properties give a natural transformation

$$\theta : \tau_{\leq m} \circ \tau_{\geq n} \to \tau_{\geq n} \circ \tau_{\leq m}$$

Proposition 2.2.2. The transformation θ is an equivalence. If $m < n$, both sides are zero; if $m \geq n$, they define the truncation of x to the interval $[n, m]$.

Proof sketch. Both sides lie in $C_{\geq n} \cap C_{\leq m}$. When $m < n$, orthogonality makes this intersection zero. When $m \geq n$, apply $\mathrm{Ext}^0_C(-, y)$ for $y \in C_{\geq n} \cap C_{\leq m}$ to the two truncation sequences. Orthogonality identifies the outer maps, and the five lemma identifies the middle map induced by θ. Yoneda then shows that θ is an equivalence. □

Definition 2.2.3 (Heart). The *heart* of the t-structure is

$$C^{\heartsuit} := C_{\geq 0} \cap C_{\leq 0}$$

Define the homology-object functors by

$$\pi_0 := \tau_{\leq 0} \circ \tau_{\geq 0} \simeq \tau_{\geq 0} \circ \tau_{\leq 0}, \quad \pi_n(x) := \pi_0(x[-n])$$

For $x, y \in C^{\heartsuit}$ and $r > 0$, orthogonality gives

$$\pi_r \mathrm{Map}_C(x, y) \simeq \mathrm{Ext}^{-r}_C(x, y) \simeq 0$$

Thus mapping spaces in $C^{\heartsuit}$ are discrete, so this ∞-category is the nerve of its homotopy category. The category $hC^{\heartsuit}$ is abelian; kernels and cokernels are obtained by applying π_0 to fibers and cofibers. Our indexing is homological: the object above is denoted $\pi_n(x)$ rather than $H_n(x)$.

2.2.3 General Localizations

Not every localization of a stable ∞-category comes from a t-structure. Given a collection S of morphisms, the localization $S^{-1}C$ is the full subcategory of S-*local* objects. An object x is S-local when, for every $f : y' \to y$ in S, composition with f induces an equivalence

$$\mathrm{Map}_C(y, x) \to \mathrm{Map}_C(y', x)$$

To compare this with a t-structure, complete f to a cofiber sequence

$$y' \to y \to y''$$

Applying $\mathrm{Ext}^i_C(-, x)$ gives

$$\ldots \to \mathrm{Ext}^i_C(y'', x) \to \mathrm{Ext}^i_C(y, x) \to \mathrm{Ext}^i_C(y', x) \to \mathrm{Ext}^{i+1}_C(y'', x) \to \ldots$$

Consequently, x is f-local exactly when

$$\mathrm{Ext}^i_C(y, x) \to \mathrm{Ext}^i_C(y', x)$$

is an isomorphism for every $i \leq 0$. In particular, locality implies $\mathrm{Ext}^i_C(y'', x) = 0$ for $i \leq 0$, while vanishing for $i \leq 1$ is sufficient for locality. If y' is zero, the exact criterion reduces simply to $\mathrm{Ext}^i_C(y, x) = 0$ for every $i \leq 0$.

2.3 Some More Propositions

2.3.1 Quasisaturated Localizations

Definition 2.3.4 (Quasisaturated Collection). Let C admit pushouts. A collection S of morphisms is *quasisaturated* if

1. every equivalence belongs to S;
2. S satisfies two-out-of-three for every composable pair;
3. a pushout of a morphism in S again belongs to S.

Intersections of quasisaturated collections are quasisaturated. Hence every collection of maps has a smallest quasisaturated closure, called the collection it *generates*.

Definition 2.3.5 (Closed under Extensions). A full subcategory $E \subset C$ is *closed under extensions* if, for every fiber–cofiber sequence $x \to y \to z$, the conditions $x, z \in E$ imply $y \in E$.

Let $L : C \to C$ be an idempotent localization, let C_L be its essential image, and let S_L be the collection of maps f for which $L(f)$ is an equivalence. The collection S_L is quasisaturated.

Proposition 2.3.3 (t-Structure Localizations). The following conditions are equivalent:

1. S_L is generated by a collection of maps of the form $0 \to a$;
2. S_L is generated by all $0 \to a$ such that $La \simeq 0$;
3. the local subcategory C_L is closed under extensions;
4. for $a \in C$ and $b \in C_L$, the map $\mathrm{Ext}^1_C(La, b) \to \mathrm{Ext}^1_C(a, b)$ is injective;
5. $C_{\geq 0} := \{a \mid La \simeq 0\}$ and $C_{\leq -1} := \{a \mid La \simeq a\}$ determine a t-structure on C.

Proof sketch. The implication (1)→(2) follows by enlarging the chosen generators to all L-acyclic objects. For (2)→(3), mapping an acyclic generator into a fiber sequence shows that an extension of local objects is local.

If (3) holds, an extension of La by a local object has local middle term; consequently an extension class which becomes split after pullback along $a \to La$ was already split. This is (4).

Under (4), the fiber sequence

$$\operatorname{fib}(a \to La) \to a \to La$$

gives the required truncation sequence. Adjunction gives orthogonality, and the injectivity in (4) supplies the remaining shifted orthogonality. Hence the two classes in (5) form a t-structure. Finally, if (5) holds, the cofiber of every map inverted by L is in $C_{\geq 0}$; pushouts and two-out-of-three show that the maps $0 \to a$ with $a \in C_{\geq 0}$ generate all of S_L. □

Thus the localizations arising as negative truncations are exactly those generated, up to quasi-saturation, by killing objects.

2.3.2 Boundedness and Completion

Definition 2.3.6 (Bounded Objects). Let C^+ consist of the objects lying in some $C_{\geq n}$, and let C^- consist of those lying in some $C_{\leq n}$. Put

$$C^b := C^+ \cap C^-$$

The t-structure is *left bounded*, *right bounded*, or *bounded* when $C = C^+$, $C = C^-$, or $C = C^b$, respectively.

Definition 2.3.7 (Left Completion). The *left completion* $\hat{C}$ is the ∞-category of compatible Postnikov towers $(x_n)_{n \in \mathbb{Z}}$ with

$$x_n \in C_{\leq n}, \quad \tau_{\leq n} x_{n+1} \simeq x_n$$

Equivalently,

$$\hat{C} \simeq \lim_n C_{\leq n}$$

The canonical functor is

$$C \to \hat{C}, \quad x \mapsto \left(\tau_{\leq n} x\right)_n$$

We call C *left complete* when this functor is an equivalence.

Proposition 2.3.4 (Basic Properties of Completion). The left completion has the following properties:

1. $\hat{C}$ is stable and inherits a t-structure;
2. the functor $C \to \hat{C}$ is exact and induces equivalences

   $$C_{\leq n} \simeq \hat{C}_{\leq n}$$

   for every n;
3. it induces an equivalence $C^+ \simeq \hat{C}^+$ on left bounded objects.

Proof sketch. Limits and cofibers of compatible towers are formed levelwise, with the shift reindexing the tower, so $\hat{C}$ is stable. The two halves of its t-structure are detected levelwise. A tower bounded above stabilizes at the corresponding truncation, which proves (2); shifting proves (3). □

In particular, left completion changes only objects which are unbounded below. Passing to the left bounded part and taking left completion are inverse constructions. Right completion and right completeness are defined dually.

Proposition 2.3.5 (Criterion for Left Completeness). Suppose that C admits countable products and that $C_{\geq 0}$ is closed under them. Then the following are equivalent:

1. C is left complete;
2. the only object in $\bigcap_{n \in \mathbb{Z}} C_{\geq n}$ is the zero object.

Proof sketch. Every countable tower has a limit, computed by the fiber sequence

$$\lim_n x_n \to \prod_n x_n \xrightarrow{\mathrm{id} - \mathrm{shift}} \prod_n x_n$$

Hence a compatible Postnikov tower has a limit in C. For every x, the cofiber of the completion map

$$x \to \lim_n \tau_{\leq n} x$$

belongs to every $C_{\geq n}$. Condition (2) therefore makes this map an equivalence. Conversely, if x belongs to every $C_{\geq n}$, then all its truncations vanish; left completeness forces $x \simeq 0$. □

2.4 Filtered Objects and Spectral Sequences

Let C be a stable ∞-category with a t-structure and heart $A \simeq C^{\heartsuit}$. A *filtered object* is a functor $x : N(\mathbb{Z}) \to C$, depicted as

$$\dots \to x_{-1} \xrightarrow{f^0} x_0 \xrightarrow{f^1} x_1 \to \dots$$

Its pth associated graded object is

$$\mathrm{gr}^p x \coloneqq \mathrm{cofib}\left(x_{p-1} \to x_p\right)$$

We now construct a spectral sequence in A beginning with

$$E_1^{p,q} \simeq \pi_{p+q} \mathrm{gr}^p x$$

2.4.1 Interval Cofibers

Definition 2.4.8 (Complex Associated to a Filtration). Adjoin a least element $-\infty$ to $\mathbb{Z}$. The filtered object x extends to an interval diagram with values $x_{i,j}$ for $-\infty \leq i \leq j$, defined by

$$x_{-\infty,j} \coloneqq x_j, \quad x_{i,j} \coloneqq \mathrm{cofib}\left(x_i \to x_j\right)$$

Thus $x_{i,i} \simeq 0$, and for $i \leq j \leq k$ there is a canonical fiber–cofiber sequence

$$x_{i,j} \to x_{i,k} \to x_{j,k} \to x_{i,j}[1]$$

Equivalently, the square with vertices $x_{i,j}, x_{i,k}, 0, x_{j,k}$ is a pushout. Conversely, every interval diagram satisfying these two properties is determined by the filtration $x_{-\infty,j}$. More precisely, restriction to the $-\infty$th row is an equivalence of ∞-categories. Hence all interval cofibers and connecting maps above are functorial and independent of choices up to a contractible space.

Remark (Underlying Chain Complex). Set

$$c_p := x_{p-1,p}[-p]$$

The connecting maps of adjacent intervals give

$$\ldots \to c_1 \to c_0 \to c_{-1} \to \ldots$$

in hC. Two consecutive maps compose to zero because they are adjacent boundary maps in a coherent pushout diagram. Thus every filtered object has a canonical chain complex of graded pieces in its triangulated homotopy category.

2.4.2 The Spectral Sequence

Proposition 2.4.6 (Spectral Sequence of a Filtered Object). For $r \geq 1$, define an object of the heart by

$$E_r^{p,q} := \operatorname{im}\Big(\pi_{p+q} x_{p-r,p} \to \pi_{p+q} x_{p-1,p+r-1}\Big)$$

The connecting maps of the interval sequences induce differentials

$$d_r : E_r^{p,q} \to E_r^{p-r,q+r-1}$$

They satisfy $d_r^2 = 0$, and there are natural isomorphisms

$$E_{r+1}^{p,q} \simeq \ker\Big(d_r : E_r^{p,q} \to E_r^{p-r,q+r-1}\Big) / \operatorname{im}\Big(d_r : E_r^{p+r,q-r+1} \to E_r^{p,q}\Big)$$

In particular, (E_r, d_r) is a spectral sequence in A, natural in the filtered object x.

Proof sketch. For $r = 1$, both terms in the defining image are $x_{p-1,p}$, giving the announced E_1-page. For general r, the differential is induced by the boundary map between the two relevant interval cofibers. The pushout identities imply that two successive boundary maps compose to zero. Exactness in the heart then identifies the image defining E_{r+1} with the kernel of d_r modulo the image of the incoming d_r. □

The differential has bidegree $(-r, r-1)$ and lowers total degree by one. On the first page one may equivalently write

$$E_1^{p,q} \simeq \pi_q c_p$$

so d_1 is simply the homology differential obtained from the underlying chain complex of graded pieces.

Remark. If C is the derived ∞-category of an abelian category and x is a filtered chain complex, this construction recovers the classical spectral sequence of a filtered complex. The interval-diagram formulation is useful because it makes every higher differential canonical and functorial.

2.4.3 Convergence

Definition 2.4.9 (Compatibility with Sequential Colimits). The t-structure on C is *compatible with sequential colimits* if C admits colimits indexed by $\mathbb{N}$ and $C_{\leq 0}$ is closed under them. Under this assumption the functors $\pi_n : C \to A$ preserve sequential colimits, and sequential colimits in the heart are exact.

Proposition 2.4.7 (Convergence for a Bounded-Below Filtration). Suppose the t-structure is compatible with sequential colimits and $x_p \simeq 0$ for all sufficiently negative p. Put

$$x_\infty := \operatorname*{colim}_p x_p, \quad a_n := \pi_n x_\infty$$

Then the spectral sequence converges to $\pi_* x_\infty$. Explicitly, for fixed (p, q) the differentials eventually vanish, and a_n has the exhaustive increasing filtration

$$F^p a_n := \operatorname{im}\big(\pi_n x_p \to a_n\big)$$

whose associated graded pieces are

$$E_\infty^{p,q} \simeq F^p a_{p+q} / F^{p-1} a_{p+q}$$

Proof sketch. The lower bound makes the incoming and outgoing intervals defining d_r eventually constant or zero, so every bidegree stabilizes. Since π_n commutes with the sequential colimit, the images of $\pi_n x_p \to \pi_n x_\infty$ form an exhaustive filtration. Applying the long exact sequence to

$$x_{p-1} \to x_p \to \mathrm{gr}^p x$$

identifies the stable page with the indicated subquotient. □

2.5 The Dold–Kan Correspondence

The classical Dold–Kan correspondence says that a simplicial object in an abelian category is exactly a chain complex concentrated in nonnegative degrees. Its stable ∞-categorical form replaces chain complexes by nonnegative filtrations; this retains all higher coherences and connects directly with the preceding spectral sequence.

2.5.1 Normalization and Denormalization

Let A be an abelian category and let $x : \Delta^{\mathrm{op}} \to A$ be a simplicial object, with face maps d_i and degeneracy maps s_i. Its *normalized chain complex* Nx is defined by

$$N_n x := \cap_{i=1}^n \ker(d_i : x_n \to x_{n-1}), \quad \partial := d_0 : N_n x \to N_{n-1} x$$

The simplicial identities give $\partial^2 = 0$. If

$$D_n x := \sum_{i=0}^{n-1} \operatorname{im}(s_i)$$

is the degenerate subobject, then there is a canonical decomposition

$$x_n \simeq N_n x \oplus D_n x$$

More generally, in an additive category the normalized piece is the image of the idempotent

$$p_n := (1 - s_{n-1} d_n) \dots (1 - s_0 d_1)$$

whenever this idempotent splits.

Definition 2.5.10 (The Dold–Kan Functor). Let $c = (c_n, \partial)$ be a chain complex in nonnegative degrees. The *Dold–Kan functor* constructs a simplicial object $\mathrm{DK}\,(c)$ whose degree-n object is

$$\mathrm{DK}(c)_n \simeq \oplus_{\eta : [n] \twoheadrightarrow [k]} c_k$$

where the sum runs over all order-preserving surjections $\eta : [n] \twoheadrightarrow [k]$. For $\alpha : [m] \to [n]$, factor

$$\eta \circ \alpha : [m] \twoheadrightarrow [\ell] \hookrightarrow [k]$$

into a surjection followed by an injection. On the η-summand, the induced structure map is the identity if the injection is the identity, is $\partial : c_k \to c_{k-1}$ if it omits only 0, and is zero otherwise. This rule defines all faces and degeneracies; the equality $\partial^2 = 0$ gives the simplicial identities.

Thus the copies indexed by nonidentity surjections freely supply exactly the degenerate simplices missing from a chain complex. For example, a complex concentrated in degree zero gives a constant simplicial object, while a complex with a in degree one and zero differential has $\mathrm{DK}\,(a[1])_n \simeq a^{\oplus n}$.

Theorem 2.5.1 (Classical Dold–Kan). For every additive category A, the functor

$$\mathrm{DK} : \mathrm{Ch}_{\geq 0}(A) \to \mathrm{Fun}\,(\Delta^{\mathrm{op}}, A)$$

is fully faithful. It is an equivalence when A is idempotent complete. In particular, for every abelian category there are inverse equivalences

$$\mathrm{DK} : \mathrm{Ch}_{\geq 0}(A) \overset{\sim}{\rightleftarrows} \mathrm{Fun}\,(\Delta^{\mathrm{op}}, A) : N$$

Proof sketch. The simplicial identities produce a normalization idempotent on x_n. Its image is $N_n x$, and splitting the complementary degeneracy idempotents yields the canonical decomposition

$$x_n \simeq \oplus_{\eta : [n] \twoheadrightarrow [k]} N_k x$$

This identifies $N\,\mathrm{DK}\,(c)$ with c and $\mathrm{DK}\,(Nx)$ with x. Additivity is enough for full faithfulness; idempotent completeness is exactly what guarantees the required splittings and hence essential surjectivity. In an abelian category the normalization can equivalently be formed by the displayed intersections of kernels. □

Corollary 2.5.2. If x is a simplicial abelian group, then its underlying simplicial set is a Kan complex and

$$\pi_n(x) \simeq H_n(Nx)$$

Thus simplicial homotopy groups become ordinary homology groups under normalization.

2.5.2 Stable Dold–Kan

For a stable ∞-category C, write

$$\mathrm{Fil}_{\geq 0}(C) \coloneqq \mathrm{Fun}\left(\mathrm{N}_\bullet(\mathbb{N}), C\right), \quad \mathrm{s}C \coloneqq \mathrm{Fun}\left(\mathrm{N}_\bullet(\Delta)^{\mathrm{op}}, C\right)$$

Objects of the first category are sequences $d_0 \to d_1 \to \ldots$, while objects of the second are simplicial objects of C.

Theorem 2.5.2 (Stable Dold–Kan). Every stable ∞-category C admits inverse equivalences

$$\mathrm{DK} : \mathrm{Fil}_{\geq 0}(C) \overset{\sim}{\rightleftarrows} \mathrm{s}C : N^{\mathrm{st}}$$

No t-structure, countable colimits, or idempotent-completeness assumption is required.

The stable normalization N^{st} sends a simplicial object x to its skeletal filtration

$$d_n \coloneqq |\mathrm{sk}_n x|$$

where $|\mathrm{sk}_n x|$ is the finite geometric realization of the n-skeleton. If $\ell_n x$ denotes the latching object, stability gives

$$\mathrm{cofib}\left(d_{n-1} \to d_n\right)[-n] \simeq \mathrm{cofib}\left(\ell_n x \to x_n\right)$$

The right-hand side is the stable normalized object in degree n. Hence the underlying chain complex of this filtration is the normalized chain complex of x in hC. The equivalence itself contains the higher null-homotopies and compatibilities which that chain complex alone forgets.

Remark (Relation with the Spectral Sequence). Suppose C has a t-structure with heart A. Applying π_q degreewise to x gives a simplicial object of A, and the first page of the spectral sequence of its skeletal filtration is

$$E_1^{p,q} \simeq N_p\left(\pi_q x\right)$$

Its first differential is the normalized differential. If the geometric realization exists and the convergence hypotheses above hold, then

$$E_1^{p,q} \Longrightarrow \pi_{p+q}|x|$$

In this sense the stable Dold–Kan equivalence is the structural source of the spectral sequence attached to a simplicial object.

3 Derived Categories

3.1 Nerves of dg-Categories

3.1.1 Introduction of the Goal

Let A be an additive category. The chain complexes in A can be organized into an ∞-category C, described informally as follows:

1. The objects of C are chain complexes in A.
2. Given $x_*, y_* \in C$, the morphisms $x_* \to y_*$ are chain maps.
3. Given $f, g : x_* \to y_*$, a 2-morphism $f \to g$ is represented by a chain homotopy, that is, maps $h_n : x_n \to y_{n+1}$ satisfying
$$d_y \circ h_n + h_{n-1} \circ d_x = f_n - g_n$$
4. …

To make this precise, we could proceed in several steps:

1. Let $\mathbf{Ch}(A)$ be the ordinary category of chain complexes in A.
2. For $x_*, y_* \in \mathbf{Ch}(A)$, form the mapping chain complex
$$\underline{\mathrm{Hom}}_A(x_*, y_*)_m := \prod_{n\in\mathbb{Z}} \mathrm{Hom}_A(x_n, y_{n+m})$$
with differential
$$(\partial f)_n := d_y f_n - (-1)^m f_{n-1} d_x$$
Its degree-zero cycles are chain maps, its degree-zero boundaries are null-homotopic maps, and
$$\mathrm{H}_m(\underline{\mathrm{Hom}}_A(x_*, y_*)) \simeq \mathrm{Hom}_{h\mathbf{Ch}(A)}(x_*, y_*[-m])$$
Composition is a chain map, so $\mathbf{Ch}(A)$ is enriched over $\mathbf{Ch}(\mathbf{Ab})$; in other words, it is a dg-category.
3. The connective truncation
$$\tau_{\geq 0} : \mathbf{Ch}(\mathbf{Ab}) \to \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$
is right-lax monoidal. Concretely, it comes with coherent maps
$$\tau_{\geq 0} m \otimes \tau_{\geq 0} n \to \tau_{\geq 0}(m \otimes n)$$
Applying it to every mapping complex preserves units and composition. Thus $\mathbf{Ch}(A)$ is enriched over nonnegatively graded chain complexes.
4. By the Dold–Kan correspondence from the preceding chapter,
$$\mathrm{DK} : \mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \to \mathrm{Fun}\left(\Delta^{\mathrm{op}}, \mathbf{Ab}\right)$$

is an equivalence. The Alexander–Whitney construction equips DK with a right-lax monoidal structure, so it again transports enrichment. The resulting simplicial abelian group of morphisms is

$$\mathrm{Map}_{\Delta}(x_*, y_*) := \mathrm{DK}\left(\tau_{\geq 0}\underline{\mathrm{Hom}}_A(x_*, y_*)\right)$$

5. Forgetting the abelian-group structure degreewise turns these mapping objects into simplicial sets. Their bilinear composition gives a simplicial category, denoted $\mathbf{Ch}_{\Delta}(A)$. Every simplicial abelian group is a Kan complex, so every mapping simplicial set of $\mathbf{Ch}_{\Delta}(A)$ is Kan; hence this simplicial category is fibrant.
6. Finally, take the homotopy coherent nerve

$$C_{\mathrm{dg}}(A) := N_{\mathrm{hc}}(\mathbf{Ch}_{\Delta}(A))$$

Since $\mathbf{Ch}_{\Delta}(A)$ is fibrant, $C_{\mathrm{dg}}(A)$ is an ∞-category. It has the same objects as $\mathbf{Ch}(A)$, while its mapping spaces retain chain maps, chain homotopies, and all higher coherent homotopies.

This procedure is indeed too complicated, so we will introduce some new concepts to simplify it.

3.1.2 What is a dg(differential graded)-category?

Definition 3.1.1 (dg-Category). Let k ba a commutative ring. A *dg-category* over k consists of the following data:

1. A collection $\{x, y, \ldots\}$ whose elements are called *objects* of C.
2. For every pair of $x, y \in C$, a *chain complex* of k-modules

$$\ldots \to \mathrm{Map}_C(x, y)_1 \to \mathrm{Map}_C(x, y)_0 \to \mathrm{Map}_C(x, y)_{-1} \to \ldots$$

which we will denoted by $\mathrm{Map}_C(x, y)_*$.
3. For every triple $x, y, z \in C$ a composition map

$$\mathrm{Map}_C(y, z)_* \otimes_k \mathrm{Map}_C(x, y)_* \to \mathrm{Map}_C(x, z)_*$$

which we can identify with a collection of k-bilinear maps

$$\circ : \mathrm{Map}_C(y, z)_p \times \mathrm{Map}_C(x, y)_q \to \mathrm{Map}_C(x, z)_{p+q}$$

satisfying the *Leibniz rule*

$$d(g \circ f) = dg \circ f + (-1)^p g \circ df$$

4. For each $x \in C$, an *identity morphism* $\mathrm{id}_x \in \mathrm{Map}_C(x, x)_0$ such that

$$g \circ \mathrm{id}_x = g, \quad \mathrm{id}_x \circ f = f$$

forall $f \in \mathrm{Map}_C(y, x)_p$ and $g \in \mathrm{Map}_C(x, y)_q$.

The composition law is required to be associative in the following sense: for every triple $f \in \mathrm{Map}_C(w, x)_p$, $g \in \mathrm{Map}_C(x, y)_q$ and $h \in \mathrm{Map}_C(y, z)_r$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f$$

In the case $k = \mathbb{Z}$, we refer to C as a *dg-category* without further qualification.